

CHANDIGARH UNIVERSITY APEX INSTITUTE OF TECHNOLOGY



EXPERIMENT 11

**Subject: Advanced Database Management
System**

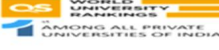
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Submitted To: Alok Sir

Question 1:

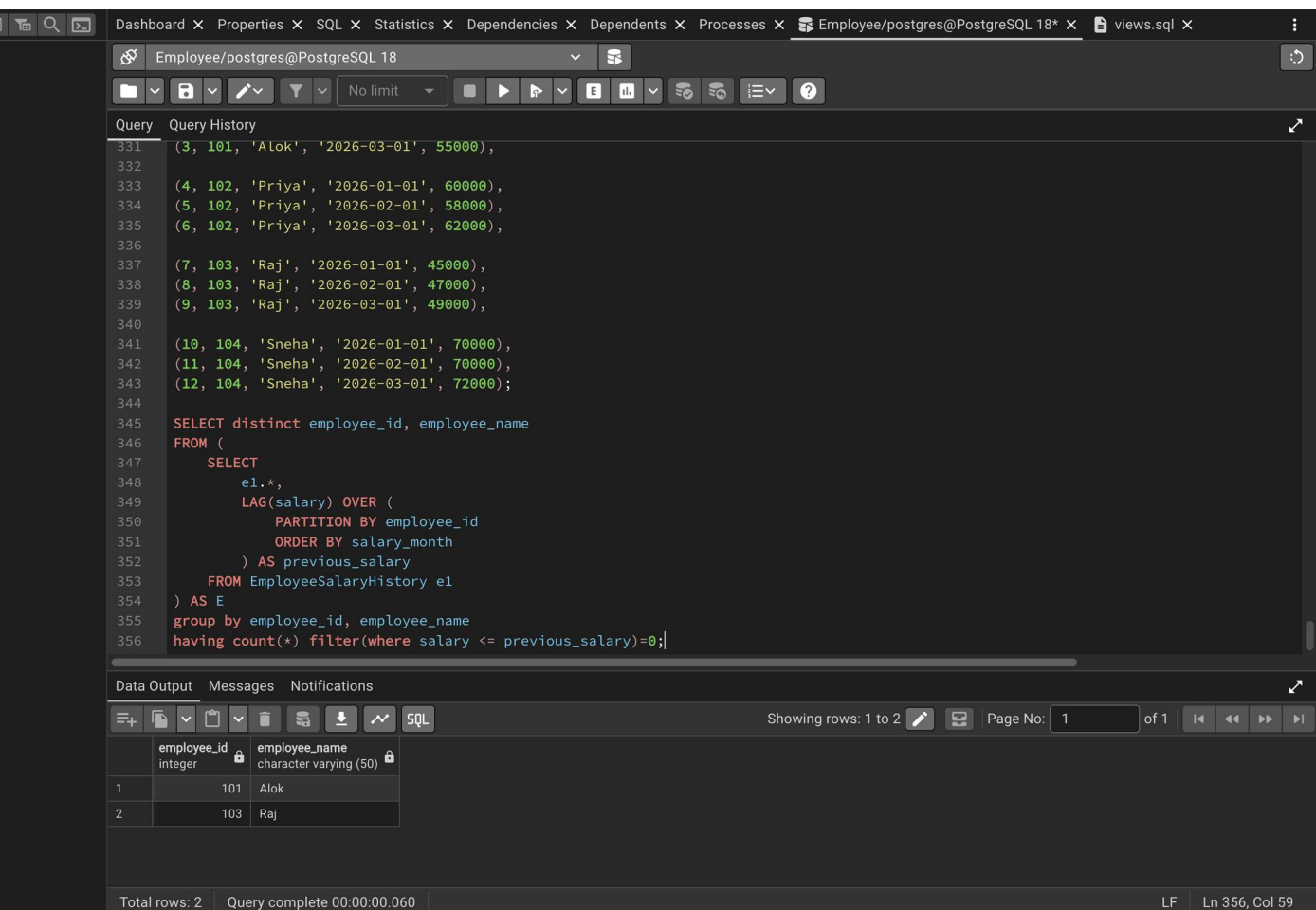
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Solution:



The screenshot shows a PostgreSQL IDE interface. The top bar includes tabs for Dashboard, Properties, SQL, Statistics, Dependencies, Dependents, Processes, and a specific connection 'Employee/postgres@PostgreSQL 18'. Below the tabs is a toolbar with various icons for file operations, execution, and help.

The main area displays a SQL query with line numbers 331 to 356. The query uses a window function to compare salaries across months for each employee.

```
331 (3, 101, 'Alok', '2026-03-01', 55000),
332
333 (4, 102, 'Priya', '2026-01-01', 60000),
334 (5, 102, 'Priya', '2026-02-01', 58000),
335 (6, 102, 'Priya', '2026-03-01', 62000),
336
337 (7, 103, 'Raj', '2026-01-01', 45000),
338 (8, 103, 'Raj', '2026-02-01', 47000),
339 (9, 103, 'Raj', '2026-03-01', 49000),
340
341 (10, 104, 'Sneha', '2026-01-01', 70000),
342 (11, 104, 'Sneha', '2026-02-01', 70000),
343 (12, 104, 'Sneha', '2026-03-01', 72000);
344
345 SELECT distinct employee_id, employee_name
346 FROM (
347     SELECT
348         e1.*,
349         LAG(salary) OVER (
350             PARTITION BY employee_id
351             ORDER BY salary_month
352         ) AS previous_salary
353     FROM EmployeeSalaryHistory e1
354 ) AS E
355 group by employee_id, employee_name
356 having count(*) filter(where salary <= previous_salary)=0;
```

Below the query editor, the 'Data Output' tab is active, showing the results of the query. It displays two columns: 'employee_id' (integer) and 'employee_name' (character varying (50)).

	employee_id	employee_name
1	101	Alok
2	103	Raj

The bottom status bar indicates 'Total rows: 2' and 'Query complete 00:00:00.060'. The page number is 1 of 1.

Question 2:

Problem Statement

[Hard]

- A company maintains employee details in an Employee table and department information in a Department table.
- Write an SQL query to find the employees whose salaries belong to the **top three distinct salary levels within their respective departments**.
- If multiple employees have the same salary, they should receive the same salary rank and all of them should be included in the result.
- The output should display:
 - Department Name
 - Employee Name
 - Salary

Solution:

```
397
398 --Solution:
399 SELECT D.name AS Department,
400        E.name AS Employee,
401        E.salary
402 FROM (
403     SELECT *,
404            DENSE_RANK() OVER (
405                PARTITION BY departmentId
406                ORDER BY salary DESC
407            ) AS rnk
408     FROM Employee
409 ) E
410 JOIN Department D
411 ON E.departmentId = D.id
412 WHERE rnk <= 3;
413
414
```

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1 of 1

	department character varying (50)	employee character varying (50)	salary integer
1	IT	Max	90000
2	IT	Jim	85000
3	IT	John	85000
4	IT	Alex	75000
5	Sales	Henry	80000
6	Sales	Susan	75000
7	Sales	David	70000
8	HR	Tina	60000
9	HR	Rita	55000